

Syllabus for MCEN 4030, Computational Methods, Spring 2003

ROOM & TIME: BESC 180, Tues. & Thurs., 5:00-6:15

INSTRUCTOR: Michael Sprague, ECME 219, 303-492-0229, Michael.Sprague@Colorado.edu
Office Hours: Tues. & Thurs., 2:30 - 3:45, ECME 215A or ECME 219 (location will depend on the number of people). While office hours are the best time for me to meet with you, you're welcome to stop by anytime; I'll let you know if I'm too busy to talk.

TA: Chung-Kyu Park, ECME 256, 303-493-6107, Chung.Park@Colorado.edu
Office Hours: Mon. & Wed., 2:00 - 3:00

TEXT: Chapra & Canale, *Numerical Methods for Engineers*, 4th edition (required)
Palm, *Introduction to Matlab 6 for Engineers* (recommended for those unfamiliar with Matlab; packaged free with new Chapra & Canale)

WEB PAGE: <http://sprocket.colorado.edu/~spraguma/MCEN4030/>
Problem sets, sample exams, and solution sets will be posted here.

GRADING:	Homework	25%	(lowest score dropped)
(Tentative)	Quizzes	10%	(lowest score dropped)
	Exam 1	20%	Thursday, February 27
	Exam 2	20%	Thursday, April 17
	Final Project	25%	

PREREQUISITES: APPM 2360 (Linear Algebra & Diff. Eq.)
GEEN 1300 (Intro Engr. Computing)

QUIZZES:

In-class pop quizzes (5-10 min.) will be given occasionally. For preparation, study the notes of the previous lecture(s) and make sure you understand the most important points. If you read the notes before coming to class, you should do fine.

E-MAIL LIST:

Please sign up for the class e-mail list *immediately*. To do this, send a message to listproc@lists with the subject blank, and the following message body: subscribe mcen4030-sp03 <your full name>. I use this list to give hints on homework, extensions, etc.

MOTIVATION & GOALS:

For many engineering positions, it is important to be able to predict or understand physical behavior through the use of mathematical models. In many of your engineering courses, you've learned how to develop models for physical systems (*e.g.*, heat transfer, fluid dynamics, system dynamics). Typically, these models are in the form of ordinary- or partial-differential equations, which are often extremely difficult, if not impossible, to solve with analytical methods (*e.g.*, separation of variables, Laplace Transforms, variation of parameters). For example, while you can likely use your existing knowledge to calculate the stress field in a beam, I'd be impressed if you could do the same for a car engine. These complicated problems are where computational or numerical methods shine.

With the rapid advancement of computers over the past several decades, numerical methods have become an invaluable tool to engineers. For example, you've likely heard of the finite-element (FE) method because it has become a standard design and analysis tool in most engineering fields.

It is not surprising that this advance in computers has been accompanied by the development of a plethora of commercial numerical software packages (ANSYS, ABAQUS, FLUENT, etc.). As a

practicing engineer, your use of, or exposure to, these canned, or black-box, packages will not be uncommon. A certain level of background or theoretical knowledge is critical to their effective use. An uninformed user will often be subjected to garbage output without knowing it!

Another advantage of a sound background in numerical methods is that you can write your own programs to solve problems. This is appealing because you can avoid paying the software-licensing fees that can easily reach the tens of thousands of dollars.

In this course, you'll be expected to implement the numerical methods discussed in class. We'll start with simple methods, which will help you get oriented with the computer and Matlab. These simple methods are often the building blocks of more complex methods. As for specific topics, we'll decide as we go along. However, I'll try to focus on methods that are in standard use today, and hopefully, we can spend plenty of time on the FE method.

COMPUTER USE & PROGRAMMING:

As the course name implies, you'll be using computers often. Further, you'll be expected to write programs. Don't worry if you haven't programmed in a while; I'll do my best to ease you back into it. The vast majority of your computer use will involve Matlab, which has a high-level programming language. Matlab may be found on many computers throughout the Engineering Center. If you know some other language (C, FORTRAN, etc.), it should be very easy to pick up Matlab's language.

Operating-system choice: You may complete your assignments and programs on the computer OS of your choice, *i.e.*, Windows, Mac, or UNIX. If you plan to go on to graduate school or believe you'll be using numerical methods often in your career, I *highly* recommend that you use a computer with UNIX. In my opinion, UNIX is the most powerful operating system for developing and using numerical methods. On the down side, first-time users find it cryptic and confusing. Sun UNIX machines are available in the SimLab. Please contact me if you'd like an account. You can readily download a free copy of UNIX (Linux) for your PC or Mac.

HOMEWORK:

Posted on the course web page each Thursday or Friday and due at 5:00 PM on the Friday of the following week. Homeworks submitted by noon on the following Monday will receive half credit. Later homework will not be credited. Homework solutions will be placed on the board across from ECME 269 and will be placed on the web page. All homework must be submitted on 8.5×11 sheets; write your name on each sheet, staple them together, and do not fold them.

Presentation: An important aspect of numerical methods is being able to effectively communicate your results with others. Thus, a portion of each homework grade will be based on the quality of your presentation. This includes content and appearance of both text and graphics. If you think you need to include raw data, put them in an appendix.

Graphs: Many of your data will be presented graphically. Please don't turn in full-page graphs unless it's warranted (very rare). Graphs should be oriented in portrait form and should be incorporated into the text when possible, and they should be numbered and labeled with a caption. When two or more data lines are in the graph, make sure that you use some type of legend. Make sure that all axes are labeled. Note that the default settings in MS Excel are about the worst I've seen. Finally, always make sure to include a discussion of the important features for each graph.

FINAL PROJECT:

We'll talk more about this later in the semester. You'll be given some freedom to pick the project; please let me know if you have any ideas for a topic.

CHEATING:

Homework and exams are individual assignments; cheating will not be tolerated. You are expected to write all programs *yourself*. Using copied programs or sections of programs in your homework constitutes cheating. However, discussing problem sets with other students is encouraged.